# Graph Attention Guidance Network With Knowledge Distillation for Semantic Segmentation of Remote Sensing Images

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***Abstract*— Deep learning has become a popular method for studying the semantic segmentation of high-resolution remote sensing images (HRRSIs). Existing methods have adopted convolutional neural networks (CNNs) to achieve better segmentation accuracy of HRRSIs, and the success of these models often depends on the model complexity and parameter quantity. However, the deployment of these models on equipment with limited resources is a significant challenge. To solve this problem, a lightweight student network framework—a graph attention guidance network (GAGNet) with knowledge distillation (KD), called GAGNet-S*—is proposed in this study, which distills knowledge from pretrained large teacher network (GAGNet-T) and builds reliable weak labels to optimize untrained student network (GAGNet-S). Inspired by the graph convolution network, this study designs a graph convolution module called the attention-graph decoder (AGD), which combines attention mechanisms with graph convolution to optimize image features and improve segmentation accuracy in the semantic segmentation task of HRRSIs. In addition, a dense cross-decoder (DCD) was designed for multiscale dense fusion, which utilizes rich semantic information in the high-level features to guide and refine the low-level features from the bottom up. Extensive experiments showed that GAGNet-S* (GAGNet-S with KD) achieved excellent segmentation performance on two widely used datasets: Potsdam and Vaihingen. The code and models are available at https://github.com/F8AoMn/GAGNet-KD.**

***Index Terms*— Dense cross-decoder (DCD), graph convolution, high-resolution remote sensing images (HRRSIs), knowledge distillation (KD), semantic segmentation.**

## I. Introduction

REMOTE sensing images (RSIs) obtained from aircraft and satellites provide a large amount of land surface information, including land surface color, texture, shape, and size. RSIs are critical for improving our understanding of active natural processes. Semantic segmentation has been widely used as a basic visual task in remote sensing [1]. Semantic segmentation classifies each pixel in an RSI into a specific semantic category, such as impervious surfaces, buildings, and trees. The high resolution and large coverage area of RSIs enable semantic segmentation to be effectively applied to the processing and analysis of large-scale surface information. RSI semantic segmentation has many important applications in urban planning [2], water resources management [3], and natural disaster monitoring [4], which provides effective support for land resource management and has great practical significance and application value.

In recent years, many researchers have adopted deep convolutional neural networks (CNNs) to fully supervise semantic segmentation [5], [6], [7]. Although significant results have been achieved, these methods require a large number of pixel-level annotations to obtain good generalizability. They improve the segmentation accuracy of a model by training it with rich and large model datasets, which are time-consuming and expensive to obtain. Therefore, model compression has become an important research topic.

Currently, large-scale high-resolution RSIs (HRRSIs) and semantic segmentation models have been overwhelmingly successful. However, deploying models with high computational complexity and multiple model parameters on resource-limited devices has become a massive challenge. To develop efficient deep-learning models, researchers have attempted to develop lightweight building blocks. Liu et al. [8] presented a lightweight network to reduce the number of model parameters. Howard et al. [9] and Sandler et al. [10] proposed MobileNets using deep separable convolution. However, because the datasets used in these studies are enormous, these lightweight methods produced poor results on the datasets. To solve this problem, knowledge distillation (KD)

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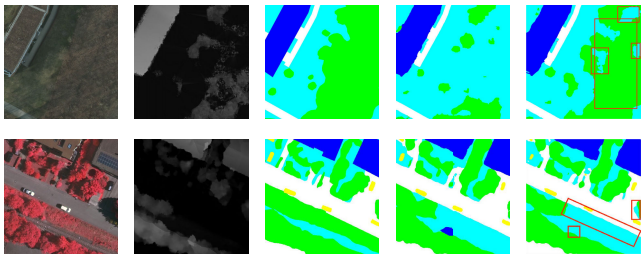



Fig. 1. Examples of problems existing in HRRSIs from the Potsdam and Vaihingen datasets: (a) RGB/IRRG, (b) nDSM, (c) ground truth (GT), (d) unimodal network, and (e) dual-modal network.

is chosen for model compression in this study. The transfer of knowledge from one model to another in KD has created a significant wave in semantic segmentation. This enables lightweight networks to improve their segmentation capability while maintaining their operational efficiency.

In addition to multispectral images, such as red-green-blue (RGB) and infrared-red-green (IRRG) images, many recent studies refer to auxiliary remote sensing data for HRRSI semantic segmentation. For example, Peng et al. [11] and Zheng et al. [12] introduced the digital surface model (DSM) corresponding to spectral images to improve the performance of semantic segmentation. DSM is a ground elevation model used for measuring the height of ground objects. This study adopted the normalized DSM (nDSM) in which the DSM height is normalized to the corresponding gray level. The gray levels corresponding to black and white are 0 and 255, respectively. The higher the altitude, the whiter the object appears in nDSM. As shown in Fig. 1, objects at higher altitudes, such as buildings and trees, tend to be whiter in nDSM. The introduction of nDSM provides rich altitude data for semantic segmentation, which offers an absolute advantage in the segmentation of high-altitude objects. To explain this, the unimodal method represented by a high-resolution context extraction network (HRCNet) [13] and the dual-modal method represented by a separation-and-aggregation gate (SA-Gate) [6] were selected in this study. As shown in Fig. 1, since the spectral features of low vegetation are similar to those of buildings, buildings are misidentified, and elevation information can be introduced to solve the corresponding problems. In this study, we focused on semantic segmentation of multispectral images and nDSM fusion.

HRRSIs are complex and special. Specifically, first, although there are differences between different types of objects, pixel features can be easily confused under the interweaving of complex backgrounds or objects. As shown in Fig. 1, pixel features of trees (green) and low vegetation (cyan) are similar, and the shadows of trees obscure other objects; therefore, it is easy to confuse pixel features in complex scenes. Second, the contour segmentation of some foreground objects is incomplete. As shown in Fig. 1, a large area of trees in the figure is not completely segmented owing to excessive scaling of objects.

CNNs have long been one of the core algorithms in the field of image recognition as well as a powerful tool for semantic segmentation tasks. However, the traditional CNN can only perform convolution on an image with a fixed size and weight, and the calculation cost is high. A graph convolutional network (GCN) solves this problem by effectively using and learning the feature relationship between each graph node and its neighbor node and by capturing the geometric features between different object regions. Therefore, this study proposes a graph-assisted dense interaction decoder, which consists of two parts: an attention-graph decoder (AGD) and a dense cross-decoder (DCD). A GCN is applied to the attention mechanism of the channel and spatial dimensions to extract context information, which enhances the semantic consistency of features. The addition of a GCN enables the model to learn the relationship between each graph node and its neighbor node in a more detailed and flexible manner, thus solving the problem of false segmentation. In addition, to further highlight the global context information and local spatial details of different levels of features, the DCD is designed for multilevel feature modeling to learn the optimal relationship between foreground and background objects, help locate objects, and completely segment object contours.

Because HRRSIs can capture a wider range of the electromagnetic spectrum from visible light to infrared (IR) and microwave (natural images can only capture the visible spectrum), HRRSIs have a larger amount of data than natural images; therefore, more powerful computational resources and more efficient algorithms are needed to process HRRSIs. In addition, the number of targets in HRRSIs is diverse, the target scale changes and the background redundancy is large, making it difficult to accurately locate objects in HRRSIs and segment complete objects. Therefore, this study introduces GCN and KD, which are widely used in natural scenes, in an attempt to gain new insights into the semantic segmentation of HRRSIs.

This study proposes a graph attention guidance network (GAGNet) with KD, called GAGNet-S*, which distills knowledge from an excellent teacher network (GAGNet-T) to optimize a lightweight student network (GAGNet-S), finding a balance between the complex architecture and precision of multimodal networks. In addition, this study proposes the AGD, an effective combination of a GCN and attention mechanisms that allows the model to flexibly capture geometric changes of different object regions and filter noise. Finally, another decoder, the DCD, uses a dense cross-interaction strategy to aggregate multilevel features, which is helpful for segmenting scenes with large-scale changes and dense complex objects.

Our primary contributions can be summarized and listed as follows:

1) We propose a KD-based graph attention guidance network, GAGNet-S*. By learning and extracting the knowledge of the existing excellent teacher network, the GAGNet-S student network is distilled to achieve an efficient segmentation effect in the multimodal HRRSI semantic segmentation task.

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2) We propose a graph-assisted dense interaction decoder, which consists of the AGD and DCD. The AGD combines a GCN and attention mechanism to adaptively guide GAGNet-S. The DCD models the multilevel feature relationships and predicts segmentation maps using the AGD.
3) We evaluated GAGNet-S* on two HRRSI benchmark datasets, and the experimental results showed that GAGNet-S* performed better than state-of-the-art (SOTA) semantic segmentation methods, and achieved optimal results in terms of the number of model parameters, model complexity, and running speed.

The remainder of this article is structured as follows. Section II describes related studies. Section III introduces the proposed GAGNet-S*. Section IV provides experimental details regarding the proposed GAGNet-S* and Section V concludes the article.

## II. Related Work

### A. Semantic Segmentation

Ronneberger et al. [14] modified and extended the architecture of fully convolutional networks (FCNs) and proposed U-Net to achieve a more accurate segmentation model trained with fewer image data. Badrinarayanan et al. [15] presented a semantic pixel-level segmentation network (SegNet) with an encoder–decoder structure based on an FCN, whose decoder performs nonlinear upsampling to generate intensive feature mapping. In addition, by constructing an encoder–decoder architecture, Chen et al. [16] extracted and combined the advantages of a spatial pyramid and depth-separable convolution to achieve a faster and stronger SegNet, DeepLabV3+. Xu et al. [13] presented HRCNet based on HRRSIs to retain spatial information and extract context information. Wang et al. [17] proposed a transformer-based decoder and constructed a UNet-like transformer (UNetFormer) for real-time urban scene segmentation. Li et al. [18] presented a multiple attention network (MANet) to extract context dependencies.

The methods mentioned above improve segmentation with the help of RGB images but require multimodal perception to obtain a reliable and comprehensive understanding of the scene under challenging real-world conditions. Extensive research has been conducted on multimodal semantic segmentation. Jiang et al. [19] presented a residual encoder–decoder network (RedNet) that used the residual module as the building block to avoid model degradation. Hu et al. [20] presented the attentional complementary network (ACNet), which preserves the flow of the original RGB-depth characteristics and utilizes fused features for feature complementarity. Sun et al. [21] proposed an RGB-thermal fusion network (RTFNet) by leveraging the advantages of thermal images. Zhou et al. [22] introduced an asymmetric three-stream self-attention network (TSNet) that effectively explores semantic structural features. Seichter et al. [23] introduced an efficient scene analysis network (ESANet), which differs from the above networks in that it uses an additional encoder to extract complementary depth geometry information to mitigate the weak segmentation effects caused by cluttered scenes. Zhang et al. [24] explored the cross-modal interaction between RGB and different complementary modes and proposed a semantic segmentation framework for RGB-X, called CMX. Yang et al. [25] focused on the influence of multilevel cue writing and context extraction on semantic segmentation and proposed a multilevel gated collaborative network (MGCNet). Ma et al. [26] used a cross-attention mechanism to integrate cross-modal information at multiple scales.

Although these methods have achieved better segmentation capabilities, they cannot overcome the limited computing resources [27]. To solve this problem, model compression has become an active research field, and KD is considered to be the most promising technology because of its excellent compression ability and performance.

### B. Knowledge Distillation

Hinton et al. [28] proposed to extract the knowledge of model integration into another unimodal and achieved good performance. Subsequently, KD was formally popularized. Wu et al. [29] proposed a communication-efficient federation learning approach based on KD that focused on cross-silo federated learning. Park et al. [30] proposed a distillation framework for self-supervision and self-training learning that enhanced the learning process of radiologists and provided better applicability in a clinical setting. KD has also been applied to semantic segmentation and has made good progress. Hu et al. [31] and Qin et al. [32] introduced KD technology into medical image segmentation, proving that student networks can effectively learn from teacher networks. Phan et al. [33] incorporated KD into the continuous learning domain of semantic segmentation, extracting the knowledge of a previous model of an old class that is similar to a new class. Ji et al. [34] presented a framework for the distillation of structural and statistical texture knowledge for semantic segmentation. Hou et al. [35] solved the problem of distilling knowledge from large teacher models to LiDAR semantic segmentation student models. Wu et al. [36] proposed a real-time semantic segmentation method via sequential KD. In addition to the above-mentioned research on KD in many fields, researchers have recently explored KD in the field of HRRSI semantic segmentation. Li et al. [37] introduced the diversity distillation loss to solve the problems of edge expansion and contraction and small object confusion. Zhang et al. [38] proposed a multigranularity semantic alignment distillation method to exploit the consistency of pixel class distribution by using pixel-level and intraclass variation distillation strategies. Dong et al. [39] used channel attention-guided feature distillation to make the two models learn complementary knowledge. Although the above methods use KD to perform HRRSI semantic segmentation tasks, they are all unimodal tasks, and the introduction of multisource information has been proved to effectively improve the recognition ability of the model [40]. Most multimodal semantic SegNets do not have real-time capability [41]. To the best of our knowledge, there is little research on KD in the multimodal semantic segmentation domain of HRRSIs. This study attempted to use KD to

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improve HRRSI segmentation accuracy and achieved fast inference while maintaining highly competitive performance in multimodal HRRSI semantic segmentation. This article proposes a lightweight student network to learn knowledge from an excellent pretrained teacher network, which solves the problems of a large number of model parameters and the poor adaptability of lightweight networks and achieves surprising segmentation performance.

### C. Graph Convolution

The proposal of the GCN has attracted wide attention, and many researchers have applied it to semantic segmentation tasks. Juarez et al. [42] proposed a network architecture combining 3-D U-Net [14] and a GCN and replaced the underlying convolutional layer of U-Net with a GCN to complete a chest CT segmentation task. Wan et al. [43] established multiscale maps to capture the spatial correlation of multiscale spectra to solve the problem of hyperspectral image classification. Wang et al. [44] iterated information fusion on video graphs by establishing fully connected graphs to capture richer relationships among video frames effectively. Shi et al. [45] integrated a GCN and deep-structured feature embedding into an end-to-end learning task to achieve better building segmentation. Liu et al. [46] designed a self-constructed graph module using a GCN to propagate contextual information and improve semantic segmentation. Xie et al. [47] proposed a scale-aware GCN to deduce the cross-scale relationship of semantically segmented images with fewer shots. Huang et al. [48] proposed a text-guided GCN for 3-D instance segmentation. Li et al. [49] used the adaptive domain GCN for inference analysis of the significance context.

By processing non-Euclidean structured data, a GCN can extract contextual information from the graph structure for reasoning and analysis. The key components of a GCN are its nodes, edges, and parameterized messaging functions, which are usually updated in an end-to-end manner. Similar to the wrong segmentation problem in HRRSIs, the flexibility of GCNs is considered of great importance in this study, which performs adaptive control through the neighborhood structure of the graph, flexibly learns the relationship between each graph node and its neighbors, and increases the difference between different categories in complex scenes, to solve problems such as incomplete segmentation. This study attempted to combine the attention mechanism along the channel and the spatial dimensions to obtain a more comprehensive semantic relationship.

## III. Proposed Methods

In this section, we first describe the overall framework of the proposed method, then separately explain the structure of each component, and finally introduce KD and the loss function adopted in this study.

### A. Overview

As shown in Fig. 2, vertically, our framework consists of GAGNet-S and GAGNet-T, as well as inputs and outputs. Horizontally, GAGNet-S consists of a backbone, fusion, enhancement strategies, and graph-assisted dense interactive decoders. The input of both GAGNet-S and GAGNet-T is an image pair of the IRRG spectral and nDSM images. The teacher network uses ResNet-50 [51] as the backbone, and GAGNet-S uses MobileNet V2 [10] as the backbone. To adapt MobileNet V2 pretrained on ImageNet to the semantic segmentation task, the last global average pooling layer and full connection layer were removed from the MobileNet V2 backbone and five outputs of different scales with different channels were obtained: $16 \times 128 \times 128$ (channel number $\times$ width $\times$ height), $24 \times 64 \times 64$, $32 \times 32 \times 32$, $160 \times 16 \times 16$, and $320 \times 8 \times 8$. Because the first-layer features contain more noise and lack semantic information [52], [53], they were removed in this study. The features extracted from the IRRG are denoted as $R_i$, $i \in \{1, 2, 3, 4\}$, and those extracted from the nDSM are denoted as $D_i$, $i \in \{1, 2, 3, 4\}$, $i$ represents the index of each stage of the network.

GAGNet-S has an encoder–decoder structure. In the encoder, eight simple enhancement blocks (SEMs) are first used to enhance the encoding features and then concatenate them to generate multimodal fusion features. The feature integration optimization block (FIOB) performs a secondary enhancement of fusion features and filters out the noise generated in the fusion process. Finally, fusion features are added to retain the original information. In the decoder, the channel number and size of the four-layer aggregation features $F_{\text{agg},i}$, $i \in \{1, 2, 3, 4\}$ generated in the encoder are adjusted with $F_{\text{agg},2}$ as the target. In this manner, four aggregation features with the same channel number and size are obtained. The decoders include the AGD and DCD. The AGD leverages the flexible feature capture ability of the GCN, adapts to different local areas of different object distributions and geometric appearances, and performs more comprehensive and deeper relational reasoning for objects of different levels. The DCD uses a dense crossing approach to process multiscale context information and combines the output features of the AGD to guide segmentation prediction maps.

### B. Fusion and Enhancement

Owing to the complexity of realistic scenes, both the IRRG and nDSM images may inevitably contain unreliable information [54]. Features may be affected by unreliable information during fusion, resulting in inaccurate segmentation prediction maps. Therefore, to address the noise interference generated during the fusion of multimodal data, the FIOB based on the self-attention mechanism is designed to filter and enhance the fusion features.

The FIOB architecture is shown in Fig. 2. The two modes are fused after SEM. SEM is composed of two groups of convolution branches aimed at extracting useful feature information in the modes to initially enhance the encoding features. The SEM processing formula is as follows:

$$\begin{aligned} S_{R,i} &= \text{Conv}_{3\times3}(\text{Conv}_{3\times3}(R_i) + \text{Conv}_{3\times3}(R_i)) \\ &\quad + \text{Conv}_{3\times3}(\text{Conv}_{3\times3}(R_i) + \text{Conv}_{3\times3}(R_i)) \\ S_{D,i} &= \text{Conv}_{3\times3}(\text{Conv}_{3\times3}(D_i) + \text{Conv}_{3\times3}(D_i)) \end{aligned} \tag{1}$$

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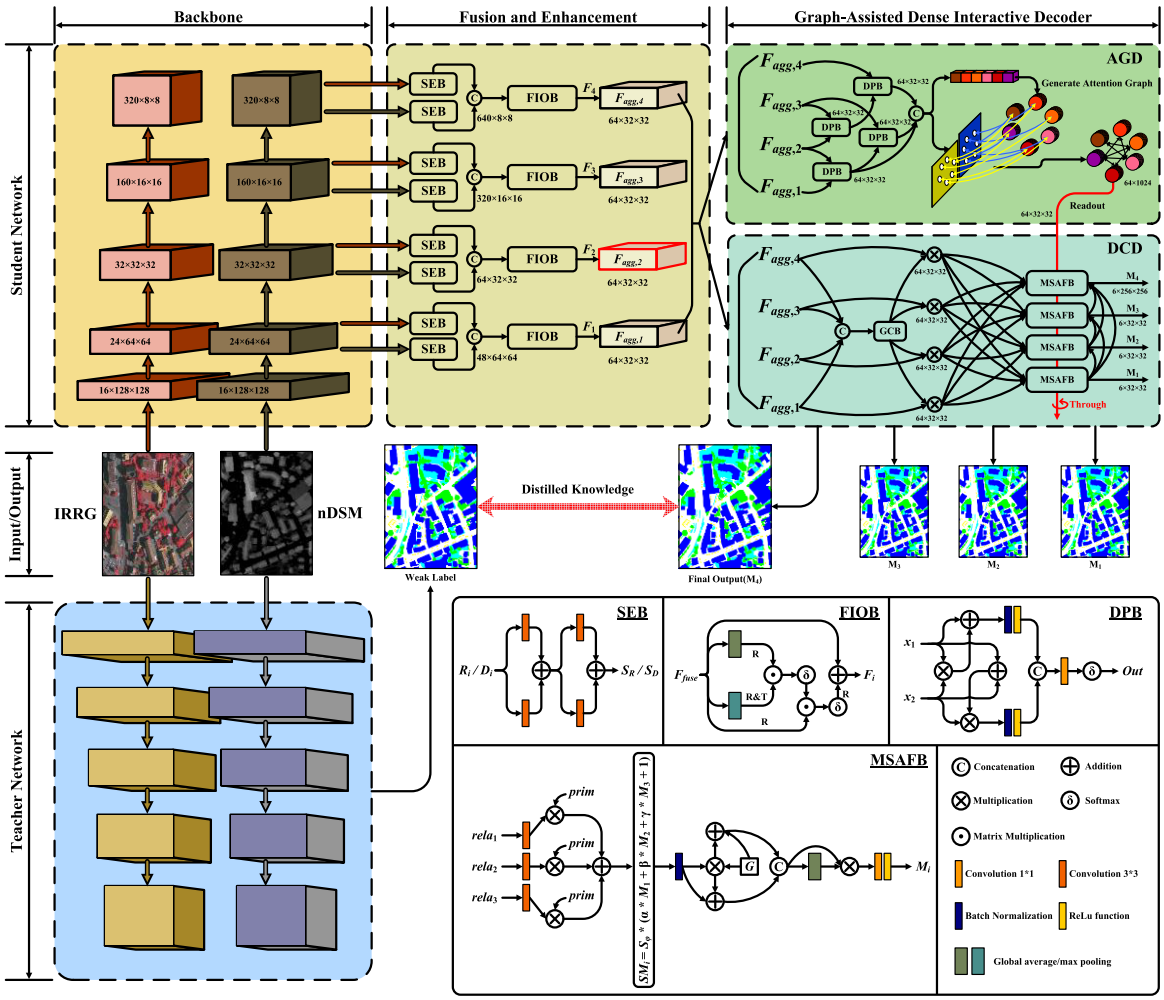



Fig. 2. Framework of our approach. In the figure, SEM represents a simple enhancement block, FIOB represents a feature integration optimization block, DPB represents a data propagating block, GCB represents a gate control block, and MSAFB represents a multilevel sensing adaptive fusion block. The teacher network adopts the network proposed by [50].

$$+\,\mathrm{Conv}_{3\times3}(\mathrm{Conv}_{3\times3}(D_i)+\mathrm{Conv}_{3\times3}(D_i)) \tag{2}$$

where $\mathrm{Conv}_{i\times i}$ represents convolution with a convolution kernel of size $i \times i$, $+$ represents the pixel-wise addition operation, and $S_{R,i}$, $i \in \{1,2,3,4\}$ and $S_{D,i}$, $i \in \{1,2,3,4\}$ represent the SEM outputs in the IRRG and nDSM branches, respectively. Next, the enhanced features of the two modes are fused to reduce modal differences and extract complementary information

$$F_{\mathrm{fuse},i} = \left[S_{R,i}, S_{D,i}\right] \tag{3}$$

where [,] represents concatenation and $F_{\mathrm{fuse},i}$, $i \in \{1,2,3,4\}$ represents fused feature.

Inspired by the nonlocal mechanism, the remote dependency relationship between channels is established by integrating the relevant features of all channel mappings to selectively emphasize the existence of interdependent channel mappings. Specifically, for the cross-modal fusion feature $F_{\mathrm{fuse},i} \in \mathbb{R}^{C\times H\times W}$, the global average pooling operation and global maximum pooling operation are adopted to make the fusion feature more sensitive to background and texture information and reduce the number of training weights to a certain extent. The two pooling features are flattened to obtain $F_{b,i} \in \mathbb{R}^{C\times N}$, $i \in \{1,2,3,4\}$ and $F_{t,i} \in \mathbb{R}^{C\times N}$, $i \in \{1,2,3,4\}$, where $N = 1$. Matrix multiplication is performed on $F_{b,i}$ and $F_{t,i}^{\mathrm{T}}$, where T represents transpose, and normalization is performed by softmax to obtain the similar matrix $F_{s,i} \in \mathbb{R}^{C\times C}$, $i \in \{1,2,3,4\}$. $F_{\mathrm{fuse},i}$ is flattened to obtain $F_{r,i} \in \mathbb{R}^{C\times N}$, $i \in \{1,2,3,4\}$, which is multiplied by the similar matrix $F_{s,i}$, and normalized by softmax to obtain the reinforcement matrix $F_{e,i} \in \mathbb{R}^{C\times N}$, $i \in \{1,2,3,4\}$; the reshaped $F_{e,i}$ is added to $F_{\mathrm{fuse},i}$ to preserve the original information. The above procedure is formulated as follows:

$$F_{b,i} = \mathrm{Flat}\left(\mathrm{Gap}\left(F_{\mathrm{fuse},i}\right)\right) \tag{4}$$

$$F_{t,i} = \mathrm{Flat}\left(\mathrm{Gmp}\left(F_{\mathrm{fuse},i}\right)\right) \tag{5}$$

$$F_{s,i} = \delta\left(\mathrm{Bmm}\left(F_{b,i}, F_{t,i}^{T}\right)\right) \tag{6}$$

$$F_{r,i} = \mathrm{Flat}\left(F_{\mathrm{fuse},i}\right) \tag{7}$$

$$F_{e,i} = \delta\left(\mathrm{Bmm}\left(F_{s,i}, F_{r,i}\right)\right) \tag{8}$$

$$F_i = F_{e,i} + F_{\mathrm{fuse},i} \tag{9}$$

where Flat represents the flattening operation, Gmp and Gap represent global maximum pooling and global average pooling operations, respectively, $\delta$ represents the softmax activation

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function, and Bmm represents matrix multiplication. In this manner, the optimized features $F_i, i \in \{1, 2, 3, 4\}$ are obtained. To facilitate decoding, this study adjusted the channel number and size of $F_i$ to be the same as $F_2$, and the adjusted features $F_i$ is referred as $F_{\text{agg},i}$:

$$F_{\text{agg},i} = \text{Align}(F_i \rightarrow F_2), \quad i \in \{1, 3, 4\} \tag{10}$$

where Align $(a \rightarrow b)$ represents the channel number and the size of $a$ is aligned with $b$ through the convolution layer and sampling.

### C. Attention-Graph Decoder (AGD)

Most existing semantic segmentation tasks [55], [56] rely on convolution kernel domain-weighted aggregation for scenario analysis. This traditional method can only be segmented under a fixed size and weight and cannot adapt to local areas with complex object distribution and large-scale changes. To adapt to a complex scene, a GCN is introduced. Recently, significant progress has been made in GCNs. GCNs can aggregate and transform the features of the neighboring nodes of each node on the graph. Therefore, GCNs can be controlled adaptively by the neighborhood structure of the graph. In addition, GCNs can handle irregular data based on non-Euclidean data, which significantly improves the fault-tolerant ability of the model. Therefore, this article uses a GCN to establish the semantic relationship between the spatial and the channel dimensions of features in the graph structure and deduces more comprehensive and deeper global information. To fully integrate the data, this article establishes intensive connections between all previous and subsequent layers to reuse features. This tightly connected structure further cross-connects features at different levels and scales and achieves a good performance in representing complex semantic relationships of multiscale objects in remote sensing scene images [57]. The data propagating block (DPB) conducts intensive connections and progressive propagation of data using cross residuals to explore the dependency relationship of neighborhood features. Specifically, first, pixel-level multiplication makes the network pay more attention to the common elements of the two features and retains the original feature information through pixel-level addition. Then, a crossover approach is adopted to obtain more comprehensive semantic information: first, two kinds of feature information are fused, and then the noise in the fused information is filtered out by multiplication. Finally, the two refined features are combined through the convolutional layer and normalized by softmax to output the final features. The formula for this procedure is as follows:

$$\begin{aligned}\text{DPB}(x_1, x_2) = \delta(\text{Conv}_{1\times1}[\mu(\eta((x_1 \times x_2) + x_1)),\\ \mu(\eta((x_1 + x_2) \times x_2))])\end{aligned} \tag{11}$$

where $\eta$ represents batch normalization, $\mu$ represents rectified linear unit (ReLU) activation, and $\times$ represents pixel-wise multiplication operation.

Four DPBs are applied to explore the dependencies of neighborhood features. Specifically, as shown in Fig. 2, first the relationship between $f_{\text{agg},1}$ and $f_{\text{agg},2}$ and $f_{\text{agg},2}$ and $f_{\text{agg},3}$ is modeled to obtain the intermediate features $X_1$ and $X_2$.

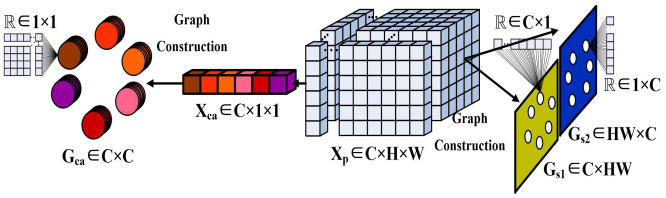



Fig. 3. Construction of channel graphs (left side) and spatial graphs (right side).

Then, feature data of $X_1$ and $X_2$ to $f_{\text{agg},3}$ and $f_{\text{agg},4}$ continue to propagate to obtain $X_3$ and $X_4$. In this manner, low-level features supplement higher-level features with detailed information, and higher-level features provide semantic cues for lower-level features. Finally, the useful information collected during the propagation process is integrated to prepare for subsequent graph decoding. The formula for this procedure is as follows:

$$X_1 = \text{DPB}\left(F_{\text{agg},1}, F_{\text{agg},2}\right) \tag{12}$$

$$X_2 = \text{DPB}\left(F_{\text{agg},2}, F_{\text{agg},3}\right) \tag{13}$$

$$X_3 = \text{DPB}\left(X_1, F_{\text{agg},3}\right) \tag{14}$$

$$X_4 = \text{DPB}\left(X_2, F_{\text{agg},4}\right) \tag{15}$$

$$X_p = \left[X_3, X_4, F_{\text{agg},1}\right] \tag{16}$$

where the function DPB$(\cdot)$ represents the operation of DPB in (11). $X_p \in \mathbb{R}^{C\times H\times W}$ represents the multilevel fusion feature obtained by DPBs.

The multilevel fusion feature $X_p$ is then decoded. The graph decoding operation is divided into graph generation in the channel and spatial dimensions. Both operations are performed in parallel.

*1) Channel Graphs:* First, the channel attention map $X_{\text{ca}} \in \mathbb{R}^{C\times1\times1}$ of $X_p$ is obtained and then structured into the channel attention graph by treating each channel of $X_{\text{ca}}$ as a vertex. The spatial information corresponding to the channel is used to describe the vertex. Therefore, $C$ vertices are obtained, and the spatial feature information of each vertex is $1 \times 1$. In this manner, the obtained $G_{\text{ca_diag}} \in \mathbb{R}^{C\times C}$ can model the semantic information between each vertex. The schematic of channel graphs construction is shown in Fig. 3.

*2) Spatial Graphs:* Feature transformation is performed on $X_p$ to obtain two forms of spatial graphs: $G_{s1}\ (V, E) \in \mathbb{R}^{C\times HW}$ and $G_{s2}\ (V, E) \in \mathbb{R}^{HW\times C}$, in preparation for the generation of the adjacency matrix. The channel information corresponding to this space is used to describe the vertices, allowing the spatial graph of the $HW$ vertices to be built. The spatial feature information of each vertex is represented by the channel feature corresponding to $\mathbb{R}^{C\times1}$. The schematic of the spatial graph construction is shown in Fig. 3. The channel matrix $G_{\text{ca_diag}}$ is then used to describe the corresponding spatial information.

To obtain the adjacency matrix $\tilde{A} \in \mathbb{R}^{HW\times HW}$, the dot product is used to compute the distance $A(i, j)$ between vertex $i$ and vertex $j$ in $\tilde{A}$. The similarity between the original feature graph $G_p \in \mathbb{R}^{C\times HW}$ and the adjacency matrix $\tilde{A}$ is then calculated to obtain the output graph $G\ (V, E)$:

$$G = \mu\left(LG_p\omega\right) \tag{17}$$

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**Algorithm 1** AGD

**Input:** The multilevel fusion feature $X_p$ obtained by DPBs;
**Output:** Graph $G(V, E)$;

1: Structure channel attention map of $X_p$:
2: $X_{ca} = tanh[Aap(X_p), Amp(X_p)]$;
3: Each channel of $X_{ca}$ is regarded as a vertex, and structures the channel attention graph $G_{ca}\ (V, E) \in \mathbb{R}^{C\times C}$;
4: Generate channel matrix $G_{ca_diag}$: $G_{ca_diag} = Bmm(G_{ca}\ (V, E), G_{ca}\ (V, E)^{\mathrm{T}})$
5: Feature transformation: two kinds of spatial graphs $G_{s1}\ (V, E) \in \mathbb{R}^{C\times HW}$ and $G_{s2}\ (V, E) \in \mathbb{R}^{HW\times C}$ are obtained by feature transformation on $X_p$;
6: Channel matrix $G_{ca_diag}$ is applied to spatial graphs $G_{s1}\ (V, E)$ and $G_{s2}\ (V, E)$ to calculate the adjacency matrix $\tilde{A} \in \mathbb{R}^{HW\times HW}$: $\tilde{A} = \delta(Bmm(Bmm(G_{ca}_diag, G_{s1}), Bmm(G_{s2}, G_{ca_diag})))$;
7: The original feature $X_p$ is reshaped to get $G_p \in \mathbb{R}^{C\times HW}$, and the similarity between the adjacency matrix $\tilde{A}$ and $G_p$ is calculated to generate the output graph $G(V, E)$;
8: **return** $G(V, E)$;

$$L = I - \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}} \tag{18}$$

where $\omega$ is a trainable weight matrix, $L$ is the graph Laplacian matrix, $\tilde{D} = \mathrm{diag}(d_1, d_2, \ldots, d_n)$, $d_i = \Sigma_j A(i, j)$, $I$ is a unit matrix, and $\tilde{A}$ is the data-dependent adjacency matrix. The pseudocode for the above program is shown in Algorithm 1, where Amp and Aap represent adaptive maximum pooling and adaptive average pooling, respectively, and tanh represents the hyperbolic tangent activation function.

Finally, the graph $G\ (V, E)$ is read out as the feature $F_G \in \mathbb{R}^{C\times H\times W}$, and then the original feature $X_p$ is added to obtain the final output of the AGD $\tilde{F}_G$:

$$F_G = \mathrm{Flat}(G) \tag{19}$$

$$\tilde{F}_G = F_G + X_p. \tag{20}$$

### D. Dense Cross Decoder (DCD)

Considering that the enhanced aggregation information can only approximately locate the target object in the scene, this study develops a dense crossover decoder to refine the aggregation information further, as shown in Fig. 2. The network learns the diversity from multilevel features, selects the most favorable feature information by element multiplication, filters unreliable noise, and more effectively integrates the bottom texture clues and top semantic clues, thus enhancing the response of the segmented objects. As described in Section III-C, DPB also adopts a dense connection approach here. This study proposes a feature adaptive dense reverse fusion mechanism, where four multilevel sensing adaptive fusion blocks (MSAFBs) are embedded in the DCD. These are used to improve the perception of multilevel features, from high to low refining features, and the refined features of the last layer are used as the final output.

Specifically, the four-layer aggregation features are fused to collect multilevel feature information, obtaining the multivariate feature $F_{\mathrm{diver}}$:

$$F_{\mathrm{diver}} = \left[F_{\mathrm{agg},1}, F_{\mathrm{agg},2}, F_{\mathrm{agg},3}, F_{\mathrm{agg},4}\right]. \tag{21}$$

The multivariate feature weight $W_G$ is then calculated via the gate control block (GCB), which consists of a convolution layer, normalization layer, and sigmoid function, as shown in (23). A weight is assigned to each layer aggregation feature $f_{\mathrm{agg},i}$ to obtain a preliminary refined feature set $\mathbb{P} = \{P_i | i = 1, 2, 3, 4\}$

$$W_G = \mathrm{GCB}(F_{\mathrm{dive}}) \tag{22}$$

$$\mathrm{GCB}(x) = \theta(\eta(\mathrm{Conv}_{1\times1}(x))) \tag{23}$$

$$P_i = W_G \times F_{\mathrm{agg},i}, \quad i \in \{1, 2, 3, 4\} \tag{24}$$

where the function GCB$(\cdot)$ represents the operation of the GCB and $\theta$ represents the sigmoid function.

The preliminary refined features $\mathbb{P} = \{P_i | i = 1, 2, 3, 4\}$ are transmitted to the MSAFBs for further adaptive feature perception enhancement. As shown in Fig. 2, on the left side of the MSAFBs, four layers of refined features are used as input, and the first extracts the important information of the other three layers of features. Subsequently, the correlation degree is calculated with the current layer feature and summed to obtain the multilevel similarity feature $S_\varphi$, as shown in lines 2–5 of Algorithm 2. On the right side of the MSAFBs, features are fused layer-by-layer according to the aggregation path from low- to high-level to obtain the multilevel fusion feature $SM_i$, as shown in lines 6–19. The output feature $\tilde{F}_G$ of the AGD throughout the four MSAFBs is used to enhance the feature adaptive sensing ability, as shown in line 20. Finally, redundant features are filtered, important information is extracted, and features are normalized to the output segmentation prediction maps $M_i$, $i \in \{1, 2, 3, 4\}$, as shown in line 21. The endpoint of the aggregation path, $M_4$, serves as the final output of the student network.

### E. Knowledge Distillation and Loss Function

Hinton et al. [28] proposed the concept of KD in 2015, which involves transferring pseudo-tags from the teacher network to the student network to improve the efficiency of the student network. In this study, we adopt this method for the HRRSIs semantic segmentation. Semantic segmentation is an intensive prediction task that assigns labels to each pixel. The structure of the student network is simple: its discrimination ability is poor; its ability to extract pixel-level features is weak, and the image content cannot be well understood in semantic segmentation tasks. In contrast, the teacher network has rich location information and strong classification abilities in the feature space. Through KD, the student network (GAGNet-S) can learn the rich pixel features and spatial information of the teacher network, significantly improving the segmentation accuracy of the GAGNet-S. In the training process of GAGNet-S*, this study explores the optimal ratio of the proportion of GT loss $a$ and the proportion of teacher network loss $b$. Numerous experiments have shown that the larger the proportion of $a$, the worse is the effect of GAGNet-S*. Finally, it can be shown that setting $a$ as 0 and $b$ as 1 can achieve the

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**Algorithm 2** MSAFB

**Input:** Preliminary refined feature set $\mathbb{P} = \{P_i | \ i = 1, 2, 3, 4\}$, the output of the previous layer of MSAFB: $M_{i-1}$, $i \in \{2, 3, 4\}$, the output of AGD: $\tilde{F}_G$;

**Output:** Segmentation prediction maps: $M_i$, $i \in \{1, 2, 3, 4\}$;

The preliminary refinement feature $\mathbb{P}$ is divided into two parts, the current layer refined feature and the other three-layer refined features. Let the former be the primary feature, named *prim* and the latter be the related feature, comprising $rela_1$, $rela_2$, $rela_3$;
A linear transformation is performed on the related features to generate a representative of the original features: $P_{rep,i} = Conv_{3\times3}(rela_i)$, $i \in \{1, 2, 3\}$;
Set the initial value of the multilevel similarity feature $S_\varphi$ to 0;
  **for** $i = 1$ to 3 **do**
    $S_\varphi = S_\varphi + P_{rep,i} \times prim$;
  **end for**
Set the initial value of $\alpha$, $\beta$, $\gamma$ to 0;
  **for** $i = 1$ to 4 **do**
    **if** $i == 1$ **then**
      **pass;**
    **else if** $i == 2$ **then**
      $\alpha = 1$;
    **else if** $i == 3$ **then**
      $\alpha = 1$;
      $\beta = 1$;
    **else**
      $\alpha = 1$;
      $\beta = 1$;
      $\gamma = 1$;
    **end if**
    $SM_i = \eta(S_\varphi \times \alpha \times M_1 + S_\varphi \times \beta \times M_2 + S_\varphi \times \gamma \times M_3 + S_\varphi)$;
    $MG_i = [(\tilde{F}_G + SM_i \times \tilde{F}_G), (SM_i + SM_i \times \tilde{F}_G)]$;
    $M_i = \mu(Conv_{1\times1}(Gap(MG_i) \times MG_i))$;
**end for**
**return** segmentation prediction maps: $M_i$, $i \in \{1, 2, 3, 4\}$;

best GAGNet-S* distillation. Therefore, the final loss $\text{Loss}_{\text{total}}$ in this article is expressed as:

$$\text{Loss}_{\text{total}} = L_{\text{CE}}\left(F^s, F^T\right) \tag{25}$$

$$L_{\text{CE}} = -\sum_{i}^{6} Y_i \times \log(C_i), \quad i \in \{0, 1, 2, 3, 4, 5\} \tag{26}$$

where $L_{\text{CE}}$ represents the extensively applied cross-entropy (CE) loss. $F^S$ and $F^T$ represent the feature maps of student and teacher networks, respectively. $Y_i$ is an indicator variable representing the $F^T$ class probability distribution of the $i$th pixel. $C_i$ represents the category probability distributions of the sample for all pixels.

## IV. Experimental Results and Analysis

### A. Datasets and Evaluation Indices

We consider two datasets for semantic segmentation of HRRSIs: the Potsdam and Vaihingen datasets from the International Society for Photogrammetry and Remote Sensing. As shown in Fig. 4, both datasets contain six common categories of urban scenes: low vegetation (cyan), buildings (blue), impermeable surfaces (white), cars (yellow), trees (green), and clutter/background (red). The Potsdam dataset contains 24 fully labeled images with a ground sampling distance of 5 cm/pixel, including five channels of near-IR, red (R), green (G), blue (B), and nDSM, with an average image size of 6000 × 6000 pixels. The multispectral images, composed of RGB bands and corresponding single-channel nDSM, are used as input data in the experiment. For the Vaihingen dataset, 16 fully labeled images with a ground sampling distance of 9 cm/pixel are included, including near-IR, red, and green 3-channel spectral images and corresponding nDSM, with an average image size of 2500 × 2500 pixels. The multispectral images, composed of IRRG and single-channel nDSM, are used as input data in this experiment.

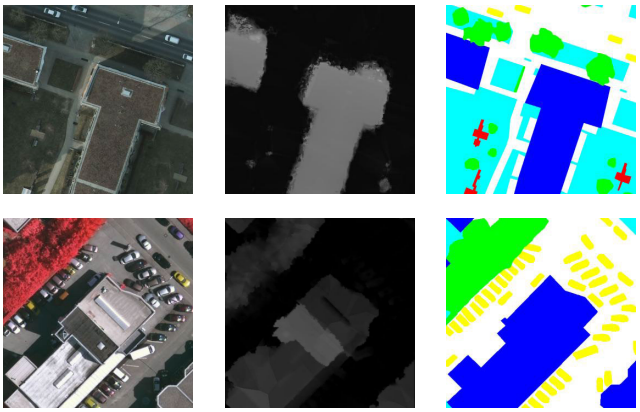


Fig. 4. Sample images from the Potsdam and Vaihingen datasets. The top images correspond with the Potsdam dataset. The bottom images correspond with the Vaihingen dataset: (a) RGB/IRRG, (b) nDSM, and (c) ground truth (GT).

Moreover, to verify the validity of our study, we adopt the evaluation indices commonly used in semantic segmentation [58]: intersection over union (IoU), class accuracy (Acc), mean IoU (mIoU), and mean Acc (mAcc) [59].

### B. Implementation Details

Because of the large image size of the original image, this study refers to Peng et al. [11]. Thus, the original dataset is clipped in the experimental data to an image size of 256 × 256 by sliding the window and dividing the two datasets into the training and test sets. For the 24 HRRSIs fully labeled in the Potsdam dataset, seven HRRSIs are selected as the test sets (ID 2_11, 2_12, 4_10, 5_11, 6_7, 7_8, and 7_10), and the remaining 17 HRRSIs are selected as the training sets. For the 16 HRRSIs fully labeled in the Vaihingen dataset, five HRRSIs are extracted and divided into the test sets (ID 11, 15, 28, 30, and 34), and the remaining 11 HRRSIs are the training sets. The same setup is adopted for all experiments in this study. Specifically, the Adam optimizer [60] is used on an NVIDIA TITAN XP 12 GB GPU workstation with a weight decay of 0.0005 and an initial learning rate of 0.0001. The learning rate is reduced to one-tenth of the original rate every

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20 epochs. The number of iterations is 200, and the batch size during the training of the student network is 36. The input image size is set to 256 × 256, and the data are enhanced by scaling, clipping, flipping, and rotation, where flipping includes horizontal and vertical flipping and rotation includes counterclockwise rotations of 90°, 180°, and 270°. In this study, the teacher network uses ResNet-50 [51], which is pretrained on ImageNet, as its backbone. Considering the large number of ResNet-50 parameters, the lightweight backbone MobileNet V2 [10] is adopted in the proposed student network.

### C. Comparisons With SOTA Models

This study compares 14 SOTA semantic segmentation methods to verify the validity of our method: FCN [5], DeepLabV3+ [16], MANet [18], UNetFormer [17], HRCNet [13], RedNet [19], ACNet [20], RTFNet [21], SA-Gate [6], TSNet [22], ESANet [23], CMANet [7], CMX [24], and MGCNet [25]. Among them, FCN, DeepLabV3+, MANet, UNetFormer, and HRCNet are unimodal semantic segmentation approaches, and multispectral images are used as the input. The other methods are RGB-depth semantic segmentation methods, and this study uses multispectral and nDSM images to replace the RGB and depth images as inputs. To prove the effectiveness of the proposed GAGNet-S*, it is compared with SOTA methods from three perspectives: quantitative, qualitative, and complexity evaluations.

*1) Quantitative Experiments:* Tables I and II list the Acc, IoU, mAcc, and mIoU obtained using the proposed method and comparison approaches on both datasets. The results show that GAGNet-S* achieves good segmentation performance. Generally, in HRRSI semantic segmentation, a model with a lightweight backbone cannot achieve good segmentation owing to the large datasets. Model size and computing speed are also challenging indicators. The teacher network has good segmentation ability in the semantic segmentation task, whereas GAGNet-S, with its lightweight backbone, does not achieve good segmentation results. Therefore, KD is an inevitable choice. Specifically, GAGNet-S is used to "imitate" the teacher network and learn its excellent feature representation and segmentation abilities. It is evident from the tables that the results of GAGNet-S* far exceed those of GAGNet-S and are close to or even exceed those of the teacher network, further proving the effectiveness of the distillation.

An impervious surface is an object with an irregular shape and large size variation. This study uses DCD, a densely intersecting module, to better aggregate the characteristic information. Therefore, good segmentation results are achieved. Due to the introduction of the elevation data, high-altitude objects, such as buildings and trees, are also well-segmented. Further, the multimodal data are optimized and enhanced, and the noise is filtered, thus improving the segmentation accuracy. Small and dense objects, such as cars, also achieve more accurate segmentation under AGD. In summary, the proposed method is superior in terms of performance on both datasets.

*2) Qualitative Experiments:* Fig. 5 shows the visualization results of semantic segmentation for the different methods, further illustrating the plasticity of the proposed method. Owing to layout limitations, we choose HRCNet as the most representative unimodal comparison method and select all multimodal comparison methods for the visual comparison. As shown in lines 3 and 6 of Fig. 5, because the spectral characteristics of the impervious surfaces and buildings are similar, the unimodal methods are prone to confusion in the process of segmentation, leading to segmentation errors, whereas the multimodal methods can correctly and clearly segment these categories because they introduce elevation data. As shown in lines 2 and 5, some multimodal methods ignore the noise interference caused during multimodal data fusion because they do not focus on noise filtering and feature enhancement, resulting in incomplete segmentation of low vegetation and buildings. Although the segmentation effect shown in the figure is similar to that achieved in some multimodal comparison methods in some scenarios, the size of the proposed model is far smaller than those of these multimodal comparison methods, which indicates that the proposed method not only pursues model accuracy but also considers the model size and further demonstrates its plasticity.

To further illustrate the effectiveness of KD, we visually compare the teacher network, GAGNet-S, and GAGNet-S*. As shown in Fig. 6, each line shows a weakness of GAGNet-S, that is, an incomplete segmentation. This is because GAGNet-S adopts the lightweight backbone (MobileNet V2) to ensure that the feature information cannot be fully extracted. The teacher network can capture high-level semantic cues and low-level spatial details to address object scale changes and spatial detail loss [50]. Under the guidance of the teacher network, GAGNet-S filters redundant information and extracts useful features of the teacher network for its own use, thereby improving the segmentation accuracy. Although the teacher and student models are not similar in structure (in the student network, the encoder uses MobileNet V2, whereas, in the teacher network, it uses ResNet-50), the teacher model is more complex than the student model, especially in terms of the number of parameters and FLOPs.

*3) Complexity Evaluation:* Table III presents the comparative results of the complexity evaluation of the different methods. We conducted a comparative experiment on the number of parameters (Params), floating-point operations (FLOPs), and computing speed (FPS) for the different methods. The traditional large-scale backbone networks VGG and ResNet are powerful, with deeper networks and more convolution cores. However, they also have problems, such as large storage of parameters and insufficient running speed. For networks using lightweight backbones, such as ESANet [23], ResNet-34, which has fewer parameters than the traditional large-scale backbone, is adopted. Although it has a low parameter number, the segmentation accuracy of the model is a major disadvantage. In this study, a lightweight backbone network (MobileNet V2) is used to effectively reduce the parameters and FLOPs. It can be observed from Table III that the number of parameters and complexity of the proposed method are considerably lower than those of the comparison methods and teacher network, and the calculation speed of the model is also significantly higher. Therefore, the proposed method is more competitive than the comparison methods in terms of the number of parameters, computational complexity, and model

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TABLE I
COMPARATIVE RESULTS OF QUANTITATIVE EVALUATION ON THE POTSDAM DATASET

| Methods | Types | Imp.surf | | Building | | Low veg. | | Tree | | Car | | Cluter | | mAcc | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Acc | IoU | Acc | IoU | Acc | IoU | Acc | IoU | Acc | IoU | Acc | IoU | | |
| DeepLabv3+[16] | Unimodal | **91.57** | 82.49 | 91.78 | 87.59 | 87.36 | **73.63** | 85.45 | 73.32 | 93.89 | 90.04 | 53.80 | **43.54** | 83.97 | 75.10 |
| FCN-8S[5] | Unimodal | 89.47 | 79.77 | 90.69 | 83.60 | 85.13 | 71.12 | 82.86 | 71.23 | 91.02 | 81.53 | 49.05 | 36.49 | 78.61 | 69.32 |
| HRCNet[13] | Unimodal | 90.03 | 81.68 | 90.87 | 85.75 | 88.17 | 73.18 | 82.02 | 71.32 | 93.94 | 89.82 | 56.72 | 40.03 | 83.63 | 73.63 |
| MANet[18] | Unimodal | 90.48 | 81.20 | 87.58 | 83.47 | **88.78** | 72.74 | 82.69 | 71.54 | 90.96 | 88.73 | 46.16 | 38.11 | 81.11 | 72.63 |
| UNetFormer[17] | Unimodal | 91.43 | 82.49 | 90.82 | 86.93 | 87.36 | **73.85** | 85.22 | **73.72** | **94.63** | 88.02 | 56.13 | 41.64 | 84.27 | 74.44 |
| RedNet[19] | Dual modal | **92.19** | **82.83** | 93.61 | **90.13** | 87.00 | 73.22 | 83.00 | 71.77 | 93.36 | 90.08 | 56.74 | **43.51** | 84.32 | **75.26** |
| TSNet[22] | Dual modal | 85.22 | 76.85 | 91.85 | 86.65 | **88.52** | 67.98 | 78.75 | 67.49 | 78.22 | 76.85 | 37.49 | 30.85 | 76.68 | 67.78 |
| SA-Gate[6] | Dual modal | 85.84 | 80.64 | 93.65 | 88.51 | 86.46 | 72.71 | 85.70 | 72.89 | 92.18 | 89.39 | **62.70** | 40.59 | **84.42** | 74.12 |
| ACNet[20] | Dual modal | 91.32 | 82.74 | 93.83 | 90.06 | 86.16 | 73.53 | **86.03** | 72.87 | 93.79 | **90.43** | 54.51 | 41.65 | 84.27 | **75.21** |
| RTFNet[21] | Dual modal | 91.46 | 81.99 | **93.84** | 89.75 | 85.41 | 73.14 | **85.93** | **73.63** | **94.18** | **90.29** | **56.78** | 40.77 | **84.60** | 74.93 |
| ESANet[23] | Dual modal | 91.38 | **82.92** | 93.69 | 89.82 | 87.10 | 73.16 | 82.48 | 70.81 | 93.08 | 88.53 | 55.68 | 43.38 | 83.90 | 74.77 |
| CMANet[7] | Dual modal | 91.75 | 81.87 | 93.64 | **90.23** | 86.42 | 73.59 | 84.70 | 72.40 | 92.74 | 89.11 | 55.97 | 42.74 | 84.20 | 74.99 |
| CMX[24] | Dual modal | 90.54 | 81.86 | 93.80 | 89.28 | 87.17 | 73.11 | 83.52 | 71.45 | 92.23 | 87.99 | 51.61 | 40.58 | 83.14 | 74.04 |
| MGCNet[25] | Dual modal | 90.61 | 81.21 | **93.87** | 89.02 | 86.72 | 72.67 | 83.70 | 72.51 | 93.39 | 88.91 | 53.24 | 39.66 | 83.59 | 74.00 |
| GAGNet-S (w/o KD) | Dual modal | 89.13 | 81.95 | 93.46 | 89.17 | 86.46 | 72.37 | 85.68 | 73.09 | **94.97** | 86.89 | 54.19 | 40.86 | 83.98 | 74.05 |
| GAGNet-S* (w KD) | Dual modal | 89.12 | **83.41** | **94.02** | 89.43 | 86.79 | **73.89** | **86.84** | **73.90** | 93.68 | 89.91 | **59.73** | **45.19** | **85.03** | **75.95** |

TABLE II
COMPARATIVE RESULTS OF QUALITATIVE EVALUATION ON THE VAIHINGEN DATASET

| Methods | Types | Imp.surf | | Building | | Low veg. | | Tree | | Car | | mAcc | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Acc | IoU | Acc | IoU | Acc | IoU | Acc | IoU | Acc | IoU | | |
| DeepLabv3+[16] | Unimodal | 90.06 | 81.11 | 87.04 | 82.7 | 76.65 | 64.44 | 88.60 | 76.64 | 42.51 | 43.10 | 76.97 | 69.49 |
| FCN-8S[5] | Unimodal | 89.66 | 79.71 | 93.22 | 86.80 | 75.83 | 64.33 | 89.22 | 75.58 | 45.12 | 40.16 | 78.61 | 69.32 |
| HRCNet[13] | Unimodal | 91.62 | 81.60 | 91.72 | 88.01 | **79.24** | **67.38** | 90.55 | 78.58 | 70.69 | 68.73 | 84.76 | 76.86 |
| MANet[18] | Unimodal | 91.57 | 83.23 | 93.77 | 90.04 | 76.29 | 65.59 | 91.91 | 77.94 | **86.03** | **81.01** | 87.91 | 79.56 |
| UNetFormer[17] | Unimodal | **92.74** | 83.49 | 92.97 | 89.58 | 76.15 | 66.19 | 92.56 | 78.66 | 82.61 | 78.58 | 87.41 | 79.30 |
| RedNet[19] | Dual modal | 91.49 | 84.62 | 94.81 | 91.07 | 78.67 | 66.59 | 91.41 | 78.27 | 59.77 | 56.06 | 83.23 | 75.32 |
| TSNet[22] | Dual modal | 87.93 | 78.98 | **95.81** | 91.47 | 71.62 | 57.03 | **94.26** | **81.26** | 67.63 | 66.86 | 83.54 | 75.12 |
| SA-Gate[6] | Dual modal | 90.99 | **85.70** | 93.85 | **91.72** | **84.95** | **68.68** | 89.06 | **79.15** | **84.27** | 78.07 | **88.62** | **80.67** |
| ACNet[20] | Dual modal | 91.95 | 85.34 | 95.45 | **91.82** | 78.64 | 66.87 | 91.20 | 78.55 | 83.12 | 76.81 | **88.07** | 79.88 |
| RTFNet[21] | Dual modal | 91.99 | 84.66 | 95.46 | 91.98 | 76.21 | 66.38 | 92.47 | 78.11 | 82.50 | **78.88** | 87.73 | **80.00** |
| ESANet[23] | Dual modal | **92.09** | 85.18 | 94.93 | 91.16 | 75.72 | 65.48 | 92.35 | 77.65 | 75.92 | 70.11 | 86.20 | 77.92 |
| CMANet[7] | Dual modal | 92.02 | 85.06 | 95.29 | 91.15 | 77.35 | 66.06 | 91.48 | 78.54 | 73.56 | 68.89 | 85.94 | 77.94 |
| CMX[24] | Dual modal | 91.78 | **85.39** | 95.15 | 91.28 | 77.00 | 66.29 | 92.59 | 78.85 | 83.23 | 78.58 | 87.95 | 80.08 |
| MGCNet[25] | Dual modal | 91.18 | 84.89 | **95.58** | 91.52 | 76.22 | 65.74 | **92.76** | 78.53 | 79.56 | 75.33 | 87.06 | 79.20 |
| GAGNet-S (w/o KD) | Dual modal | 90.79 | 84.35 | **96.20** | **92.48** | **79.93** | 67.11 | 90.38 | 78.41 | 78.62 | 73.51 | 87.19 | 79.17 |
| GAGNet-S* (w KD) | Dual modal | 91.38 | **85.80** | **96.59** | **92.04** | **80.46** | **68.63** | 91.17 | **79.69** | **84.63** | 78.01 | **88.85** | **80.84** |

reasoning speed, with more practical applicability. In addition, KD is used in this study to offset the poor performance of the lightweight backbone segmentation.

### D. Comparative Experiments on the UDD6 Dataset

UDD6 [61], which is an urban unmanned aerial vehicle (UAV) dataset, comprises 141 images with an image size of 4000 × 3000 or 4096 × 2160; among these, 106 images are selected as the training set and 35 images as the validation set. The UAV aerial images included in this single source dataset are RGB three-channel color images, which are divided into six public categories: other, facade, road, vegetation, vehicle, and roof.

The comparison results of GAGNet-S on UDD6 are shown in Table IV. In addition to the three benchmark methods associated with the official dataset, this study selects two excellent unimodal comparison methods, UNetFormer [17] and HRCNet [13]. The segmentation effect of the proposed GAGNet-S is superior on UDD6 in terms of the two evaluation indicators, i.e., overall accuracy (OA) and mIoU.

[1] [Online]. Available: https://github.com/MarcWong/UDD

### E. Ablation Studies

Ablation experiments are conducted to verify the validity and robustness of each component in the proposed student network. The parameters of the ablation experiment are set to be the same as those of the complete GAGNet-S* experiment. Tables V and VI list the setting details and quantitative results of the current popular ablation methods, where the symbol √ indicates that the corresponding module is reserved. Quantitatively, it can be concluded from Tables V and VI that each module helps GAGNet-S* improve segmentation accuracy to

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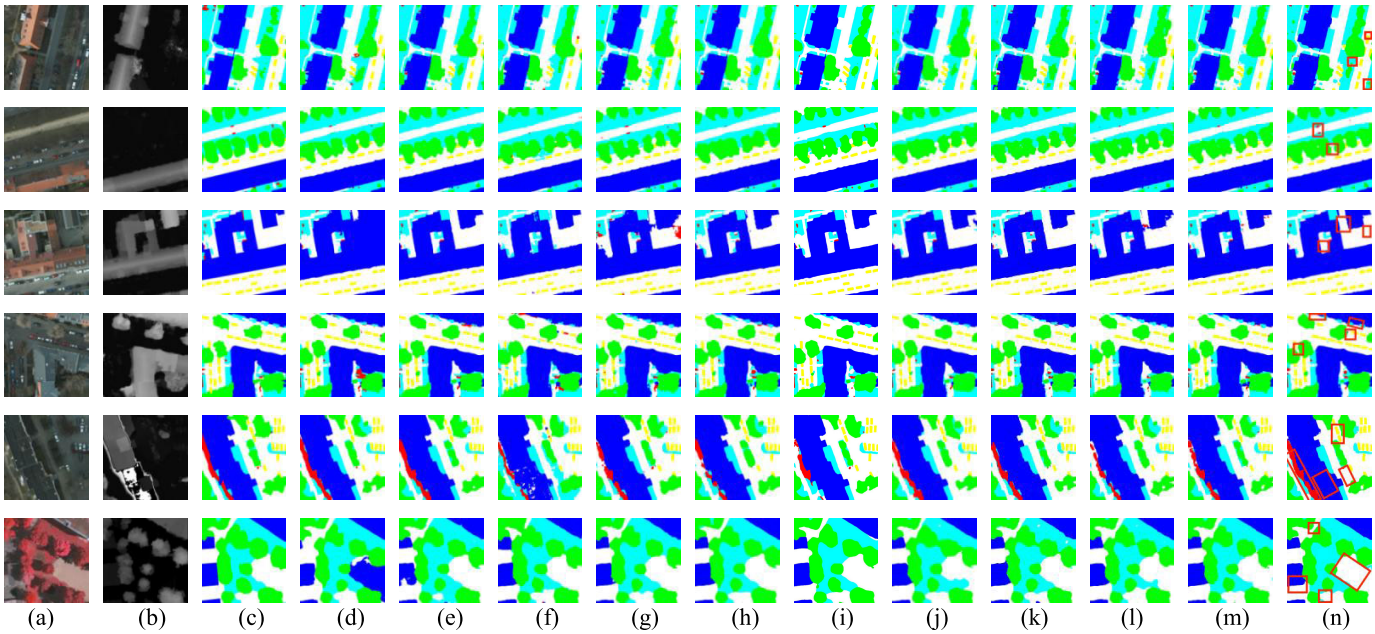



Fig. 5. Semantic segmentation maps obtained from GAGNet-S* and comparative methods: (a) RGB/IRRG, (b) nDSM, (c) GT, (d) HRCNet, (e) RedNet, (f) TSNet, (g) SA-Gate, (h) ACNet, (i) RTFNet, (j) ESANet, (k) CMANet, (l) CMX, (m) MGCNet, and (n) GAGNet-S*.

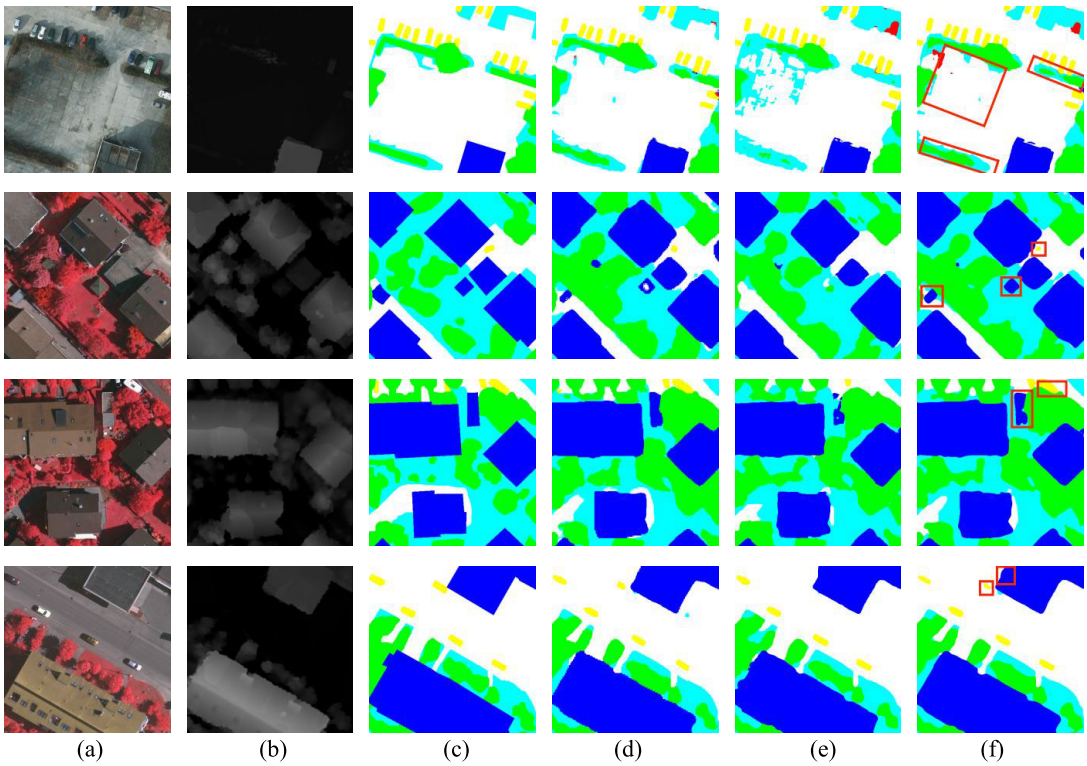



Fig. 6. Visualization maps of the effectiveness of GAGNet-S*: (a) RGB/IRRG, (b) nDSM, (c) GT, (d) Teacher, (e) GAGNet-S, and (f) GAGNet-S*.

varying degrees. Based on the backbone, whether each module is added separately or one by one, the model segmentation effect is improved, thus verifying the effectiveness of each module. Moreover, FIOB does not contribute to this method but is intended to fuse the two modes. The results of the ablation experiment prove the necessity of fusing multimodal information. Next, the qualitative analysis of AGD and DCD is presented.

*1) Effectiveness of AGD:* To study the ablation of AGD, we adopt two ablation methods; the settings are presented in Tables V and VI. The first method directly uses $F_{\text{fuse},i}$, $i \in \{1, 2, 3, 4\}$ as the AGD input based on the backbone; it deletes the other modules and then adds $F_{\text{fuse},i}$ and the AGD output as the final model output. The second model compares the effects of adding only FIOB and both FIOB and AGD to the backbone of the models. As shown in each line of Fig. 7, the location of

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TABLE III
COMPARATIVE RESULTS OF COMPLEXITY EVALUATION OF DIFFERENT METHODS

| Methods | Backbones | Params (M)↓ | FLOPs (GMac)↓ | FPS (f/s)↑ |
|---|---|---|---|---|
| DeepLabv3+[16] | ResNet-101 | 32.45 | 59.33 | 45.78 |
| FCN-8S[5] | VGG-16 | 74.55 | 134.29 | 51.5 |
| HRCNet[13] | ResNet-50 | 62.71 | 30.27 | 17.91 |
| MANet[18] | ResNet-50 | 19.49 | 35.86 | 37.45 |
| UNetFormer[17] | ResNet-50 | 24.24 | **6.03** | **68.67** |
| RedNet[19] | ResNet-50 | 21.17 | 81.95 | 21.68 |
| TSNet[22] | ResNet-34&VGG-16 | 34.27 | 41.8 | 17.91 |
| SA-Gate[6] | ResNet-101 | 41.23 | 110.85 | 22.89 |
| ACNet[20] | ResNet-50 | 26.41 | 116.6 | 16.48 |
| RTFNet[21] | ResNet-50 | 185.24 | 52.29 | 24.24 |
| ESANet[23] | ResNet-34 | **10.15** | 45.42 | 17.91 |
| CMANet[7] | ResNet-50 | 117.82 | 28.61 | 12.12 |
| CMX[24] | SegFormer | 14.26 | 66.56 | 31.69 |
| MGCNet[25] | ConvNeXt | 47.51 | 209.99 | 10.84 |
| Teacher | ResNet-50 | 189.32 | 50.92 | 19.62 |
| GAGNet-S/ GAGNet-S* | MobileNet V2 | **3.19** | **16.97** | **58.86** |

TABLE IV
COMPARATIVE RESULTS OF QUALITATIVE EVALUATION ON THE UDD6 DATASET

| Method | OA | mIoU |
|---|---|---|
| GCN[1] | 86.54 | 72.58 |
| ENet[1] | 85.20 | 69.44 |
| DeepLabv3+[1] | 86.90 | 73.18 |
| HRCNet[13] | 86.33 | 71.42 |
| UNetFormer[17] | 87.57 | 74.03 |
| GAGNet | **87.75** | **74.51** |

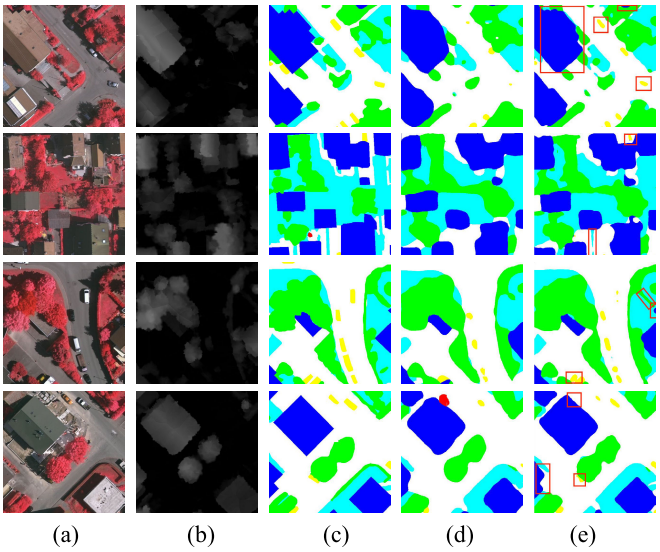



Fig. 7. Visualization maps of the effectiveness of AGD: (a) IRRG, (b) nDSM, (c) GT, (d) backbone, and (e) backbone with AGD.

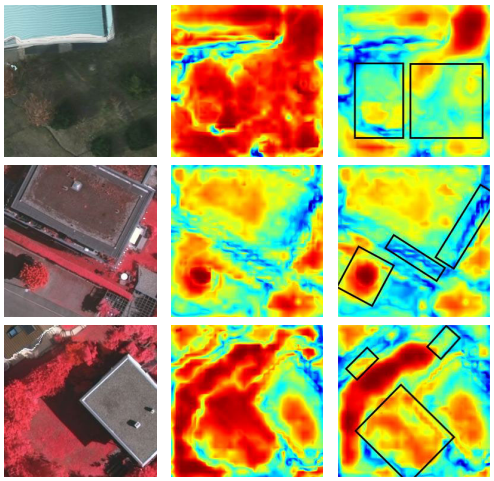
TABLE V
ABLATION STUDY OF EACH MODULE ON THE POTSDAM AND VAIHINGEN DATASETS

| Backbone | FIOB | AGD | DCD | Potsdam | | Vaihingen | |
|---|---|---|---|---|---|---|---|
| | | | | mAcc | mIoU | mAcc | mIoU |
| √ | | | | 80.49 | 71.23 | 78.30 | 70.02 |
| √ | √ | | | 81.18 | 71.54 | 79.26 | 71.05 |
| √ | | √ | | 83.69 | 74.29 | 86.68 | 78.87 |
| √ | | | √ | 83.67 | 74.77 | 85.99 | 77.50 |
| √ | √ | √ | √ | **85.03** | **75.95** | **88.85** | **80.84** |

small target objects, such as cars, is more accurate after adding AGD. Moreover, in lines 1 and 4, the contours of buildings are more clearly divided. In addition, AGD addresses the problem of segmentation errors; for example, the building location is accurately located in lines 1 and 3, and the low vegetation, impervious surfaces, and trees—which are of low altitude and have similar pixel features—are not confused, as shown in lines 2 and 3. Typically, the decoder uses high-level semantic information extracted by the encoder to locate the target position and gradually recovers the image resolution and target spatial details. AGD can help the proposed model avoid the above situations using the powerful image spatial exploration ability of GCN and improve the segmentation accuracy of the model. In order for readers to see the role of AGD more intuitively, we show the feature visualization before and after AGD decoding in Fig. 8. It can be seen that after implementing AGD is completed, the contours between multiple objects and within the same object are clearer and more complete, and objects can be located more accurately. Specifically, as shown in line 1, there is a clear distinction between trees and ground after implementing AGD. In line 2, the positioning of trees after AGD is more accurate, and the outline of the building is clearer. In line 3, because the pixel characteristics of trees and low vegetation are almost the same, the two can be better distinguished after AGD. The graph decoding method proposed in this study includes channel and spatial dimensions. In order to verify their effectiveness, ablation studies are conducted. As shown in Table VII, the segmentation effect of decoding using only channel or spatial graphs is not as good as that obtained combining both. Further ablation experiments are conducted to evaluate the influence of DPB. Table VIII demonstrates that the segmentation results without DPB (WO/DPB) are not as good as the overall model results. By cross-linking features of different levels and scales, DPB enables the model to learn the dependencies of domain features better and lays a solid foundation for subsequent map decoding.

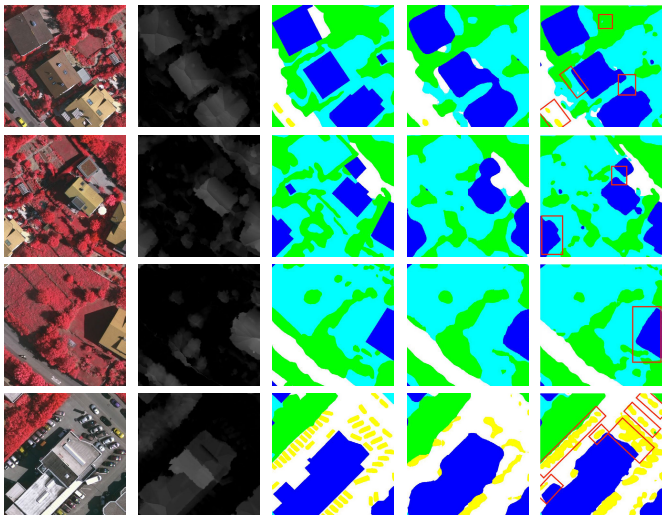
*2) Effectiveness of DCD:* Similar to that for AGD, we adopt two ablation methods for the ablation of DCD; the set-specifics are presented in Tables V and VI. The first method directly uses $F_{\text{fuse},i}$, $i \in \{1, 2, 3, 4\}$ as the DCD input based on the backbone and modifies the MSAFB input, that is, it removes the AGD output. Subsequently, the remaining modules are removed, and the output of the DCD is used as the final model output. The second method compares the overall model with the one that adds FIOB and AGD based on the backbone. As shown in lines 1, 3, and 4 in Fig. 9, small target objects,

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(a) (b) (c)

Fig. 8. Visualization of features in AGD. (a) RGB/IRRG. (b) Features fed into AGD. (c) Features generated by AGD.



(a) (b) (c) (d) (e)

Fig. 9. Visualization maps of the effectiveness of DCD: (a) IRRG, (b) nDSM, (c) GT, (d) backbone, and (e) backbone with DCD.

such as cars, are not easily ignored under the guidance of DCD. In line 1, after adding DCD, objects such as trees with large size changes and irregular shapes can be accurately classified; moreover; low vegetation at low altitudes under the shadow of high-altitude buildings is not confused with impervious surfaces. In line 4, the densely packed cars are segmented more completely using DCD. In addition, the edge division effect of buildings in lines 1, 2, and 4 is significantly improved; the problem of undivided buildings can be avoided, and the building contour is clearer. DCD densely integrates multilevel aggregation of features, resulting in a superior ability to segment dense and small objects and those with large size variations. In addition, the results of the ablation experiment of MSAFB in Table VIII confirm that the segmentation results without MSAFB (WO/MSAFB) are inferior to the overall model results. The four MSAFBs embedded in DCD improve the model perception of multilevel features and act as a "bridge" between AGD and DCD.

TABLE VI

ABLATION STUDY OF EACH MODULE AGGREGATED ONE-BY-ONE ON THE POTSDAM AND VAIHINGEN DATASETS

| Backbone | FIOB | AGD | DCD | Potsdam | | Vaihingen | |
|---|---|---|---|---|---|---|---|
| | | | | mAcc | mIoU | mAcc | mIoU |
| √ | | | | 80.49 | 71.23 | 78.30 | 70.02 |
| √ | √ | | | 81.18 | 71.54 | 79.26 | 71.05 |
| √ | √ | √ | | 84.03 | 74.48 | 87.34 | 79.50 |
| √ | √ | √ | √ | **85.03** | **75.95** | **88.85** | **80.84** |

TABLE VII

ABLATION STUDY OF GRAPH DECODING ON THE POTSDAM AND VAIHINGEN DATASETS

| Channel Graph Decoding | Spatial Graph Decoding | Potsdam | | Vaihingen | |
|---|---|---|---|---|---|
| | | mAcc | mIoU | mAcc | mIoU |
| √ | | 84.36 | 74.11 | 88.07 | 79.87 |
| | √ | 84.34 | 74.26 | 86.00 | 76.68 |
| √ | √ | **85.03** | **75.95** | **88.85** | **80.84** |

TABLE VIII

ABLATION STUDY OF DPB AND MSAFB ON THE POTSDAM AND VAIHINGEN DATASETS

| | Potsdam | | Vaihingen | |
|---|---|---|---|---|
| | mAcc | mIoU | mAcc | mIoU |
| WO/DPB | 84.11 | 75.07 | 88.26 | 79.60 |
| WO/MSAFB | 84.40 | 75.60 | 88.24 | 80.45 |
| GAGNet-S* | **85.03** | **75.95** | **88.85** | **80.84** |

TABLE IX

EXPERIMENTAL RESULTS OF UNIMODAL SEGMENTATION ON THE POTSDAM AND VAIHINGEN DATASETS

| | Potsdam | | Vaihingen | |
|---|---|---|---|---|
| | mAcc | mIoU | mAcc | mIoU |
| IRRG+IRRG | 83.15 | 74.05 | 87.00 | 78.79 |
| nDSM+nDSM | 58.00 | 45.10 | 63.06 | 52.15 |
| GAGNet-S* | **85.03** | **75.95** | **88.85** | **80.84** |

*3) Effectiveness of Dual-Modal Ablation Study:* For the dual-modal ablation study, this study replaces the original modals of IRRG and nDSM with the modals of IRRG + IRRG or nDSM + nDSM, respectively. As shown in Table IX, unimodal segmentation has obvious disadvantages compared to dual-modal segmentation, which provides complementary information to improve the segmentation effect.

*4) Effectiveness of the Number of Layers:* For the ablation study on the number of layers of the input features at the decoding stage, two different experiments are conducted. In the first experiment, the last three layers ($f_{\text{agg},2}$, $f_{\text{agg},3}$, $f_{\text{agg},4}$) are considered for decoding, while in the second experiment, only the last two layers ($f_{\text{agg},3}$, $f_{\text{agg},4}$) are considered. As shown in Table X, due to the lack of detailed information provided by the low-layer features and decline of feature interactivity in AGD and DCD, the segmentation is

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TABLE X
EXPERIMENTAL RESULTS OF DIFFERENT NUMBER OF LAYERS ON THE POTSDAM AND VAIHINGEN DATASETS

| | Potsdam | | Vaihingen | |
|---|---|---|---|---|
| | mAcc | mIoU | mAcc | mIoU |
| Last 3 layer | 84.00 | 74.74 | 86.54 | 78.36 |
| Last 2 layer | 83.88 | 73.93 | 86.43 | 78.21 |
| GAGNet-S* | **85.03** | **75.95** | **88.85** | **80.84** |

not comprehensive and sufficiently detailed; thus, the segmentation effect is significantly decreased.

## V. CONCLUSION

A semantic segmentation method based on KD and GCN was proposed for HRRSI scenes to save operational costs without sacrificing performance. A lightweight student network, GAGNet-S, "imitates" the heavyweight teacher network. The predicted result of the teacher network is regarded as a weak label to supervise the untrained GAGNet-S, overcoming the limitation of pixel-level labeling images and improving the generalizability of the lightweight model. Extensive experiments on two public datasets showed that compared with 14 SOTA methods, GAGNet-S* has absolute advantages in terms of IoU, mIoU, Acc, and mAcc. The performance of GAGNet-S* rapidly improved when using KD. On the Potsdam dataset, mAcc and mIoU improved by 1.25% and 2.57%, respectively, compared with the original student network. On the Vaihingen dataset, mAcc and mIoU improved by 1.90% and 2.11%, respectively, compared with the original student network. In addition, this study compared the model complexity of the 14 SOTA methods and the teacher network. GAGNet-S had the least parameters and low complexity, with a calculation speed that is 14.29% faster than that of FCN. These experimental results confirm that introducing the KD compression method results in a win-win situation in terms of performance and operation cost.

There are limitations to our framework. First, the generalizability of the framework needs to be improved. Specifically, model performance is affected when changing a scene image for segmentation. Therefore, data enhancement, regularization, and other methods can be adopted to improve the generalizability of the model. Second, the segmentation effect on object edges is not ideal; therefore, we can increase the loss function of boundary segmentation and separate the modeling and learning of boundary features and internal features to improve the effect of boundary segmentation.

In future work, we will explore combinations with other efficient HRRSI semantic segmentation methods to further reduce computational costs. In addition, we intend to deploy the proposed framework in areas, such as autonomous driving and robotic vision to learn from intelligent user analysis models to better advance the field of HRRSI semantic segmentation.

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